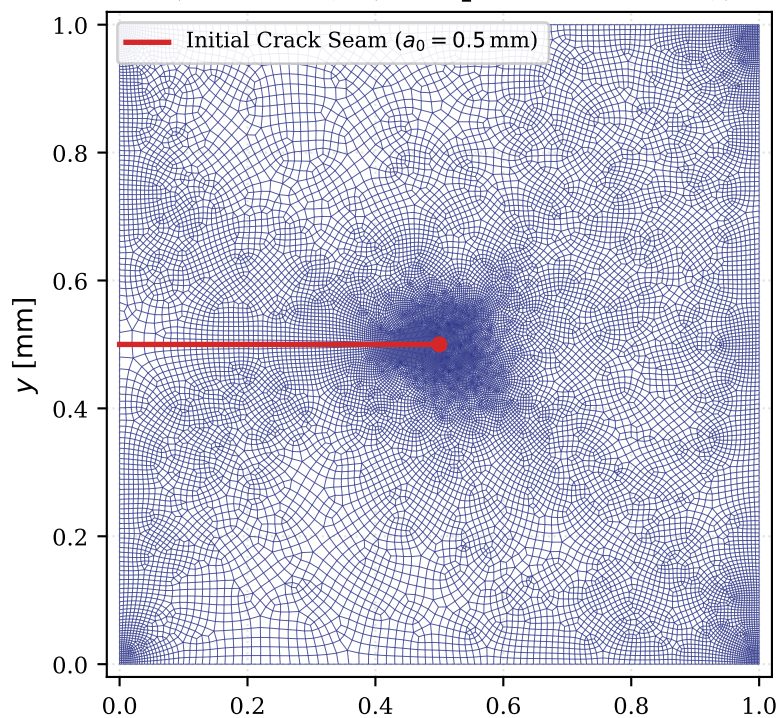
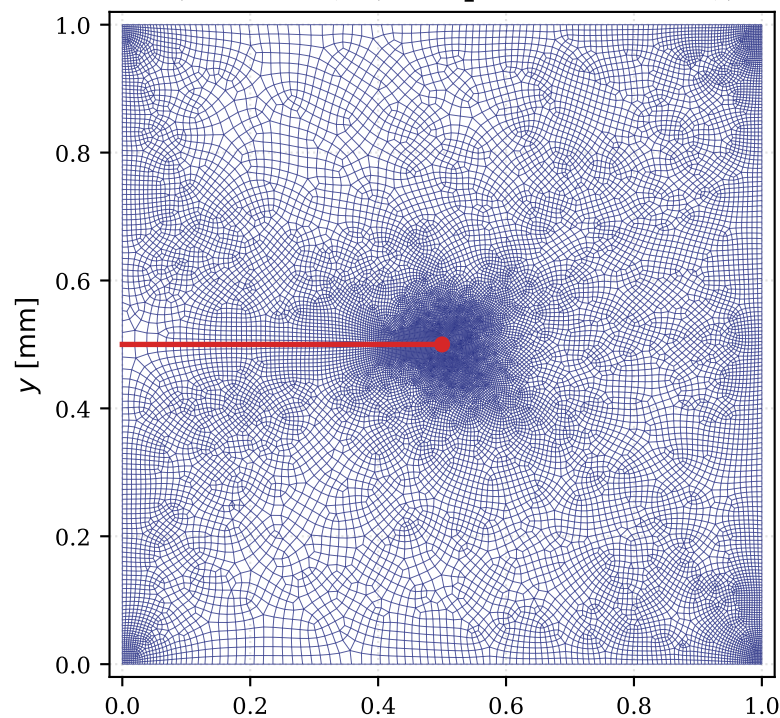


Mode-II Native Adaptive Mesh: Step-1 vs Step-2 Final Frame Side-by-Side Comparison
(Source ODB: Job-1_UEL_paper_horizon. odb, Rule: MISESeri_Adaptive_Rule, errorTarget = 2.0%)

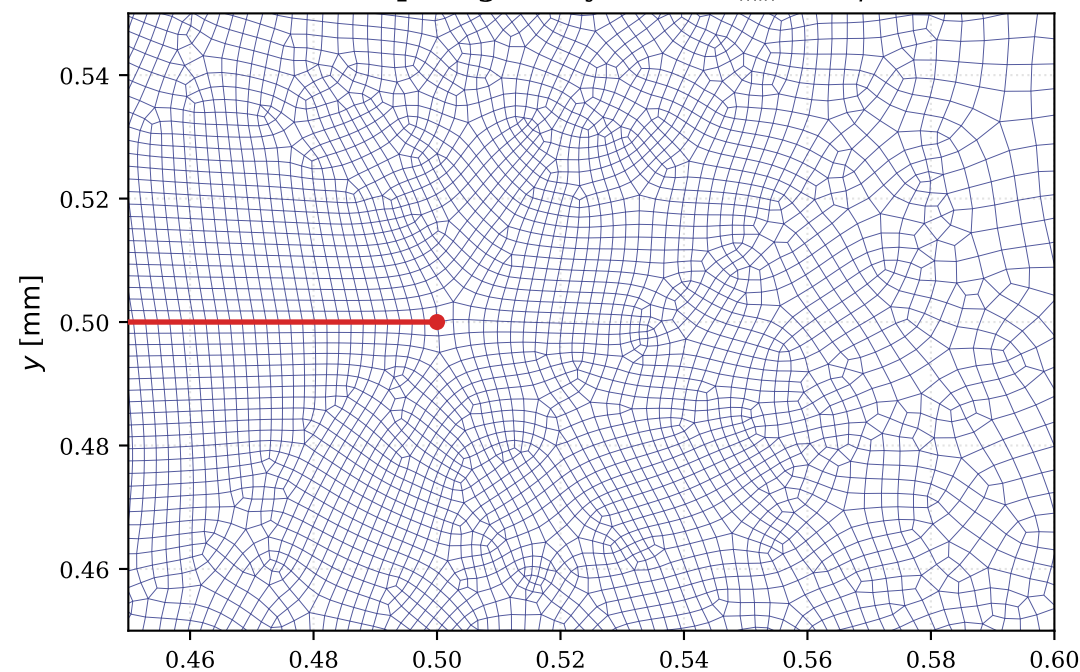
Step-1 Final Frame ($u_x = 10\mu\text{m}$)
Full Domain: 22,530 FEs (21,962 quads + 568 tris), 22,642 nodes



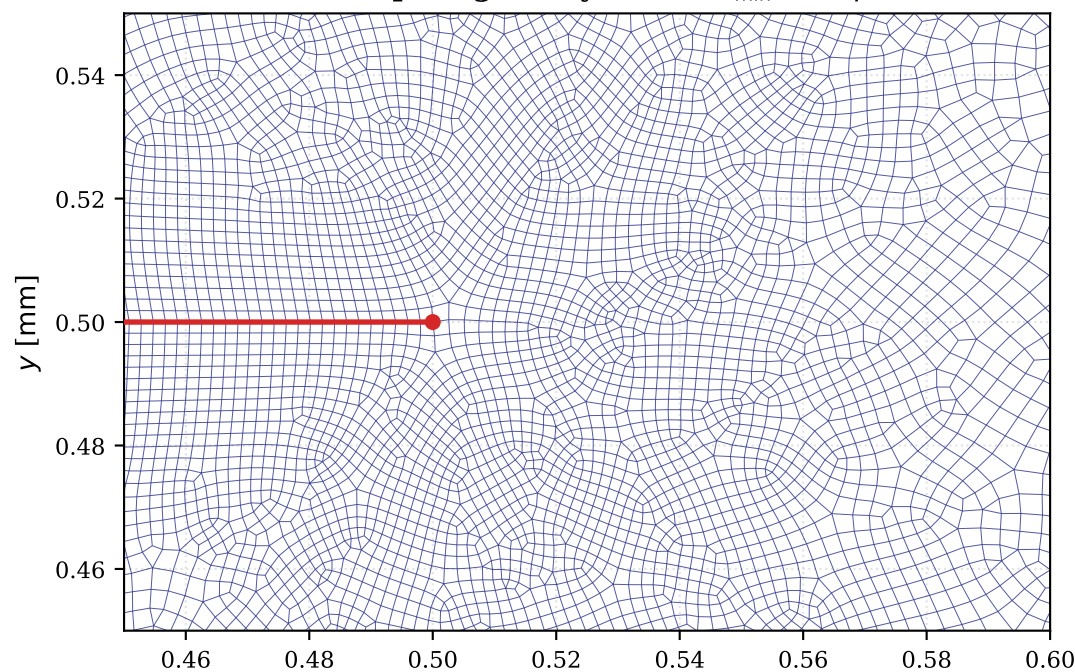
Step-2 Final Frame ($u_x = 20\mu\text{m}$)
Full Domain: 22,405 FEs (21,827 quads + 578 tris), 22,512 nodes



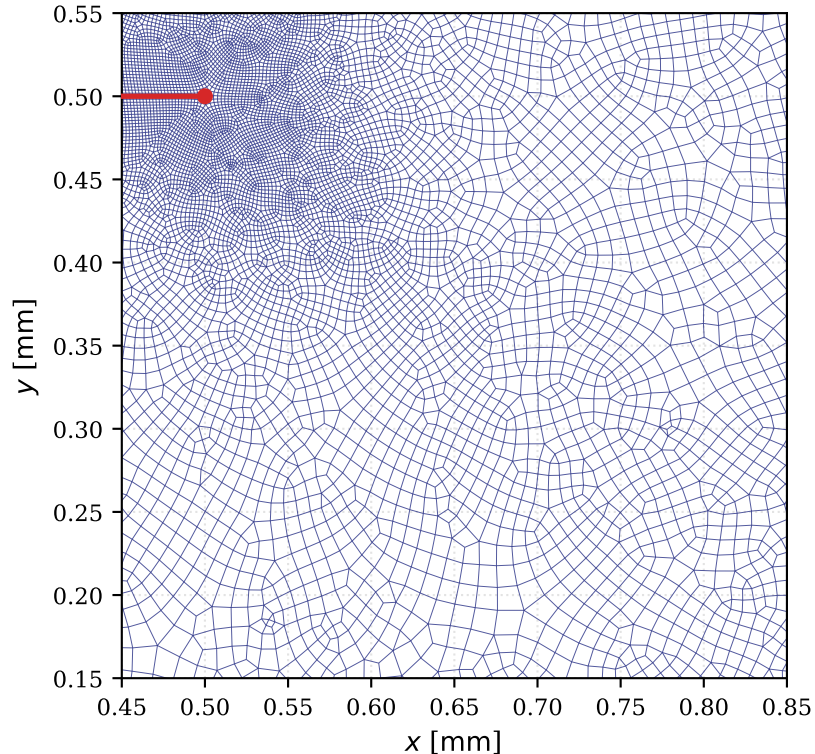
Step-1 Final Frame ($u_x = 10\mu\text{m}$)
Crack-Tip Singularity Zoom [$h_{\min} = 1.0\mu\text{m}$]



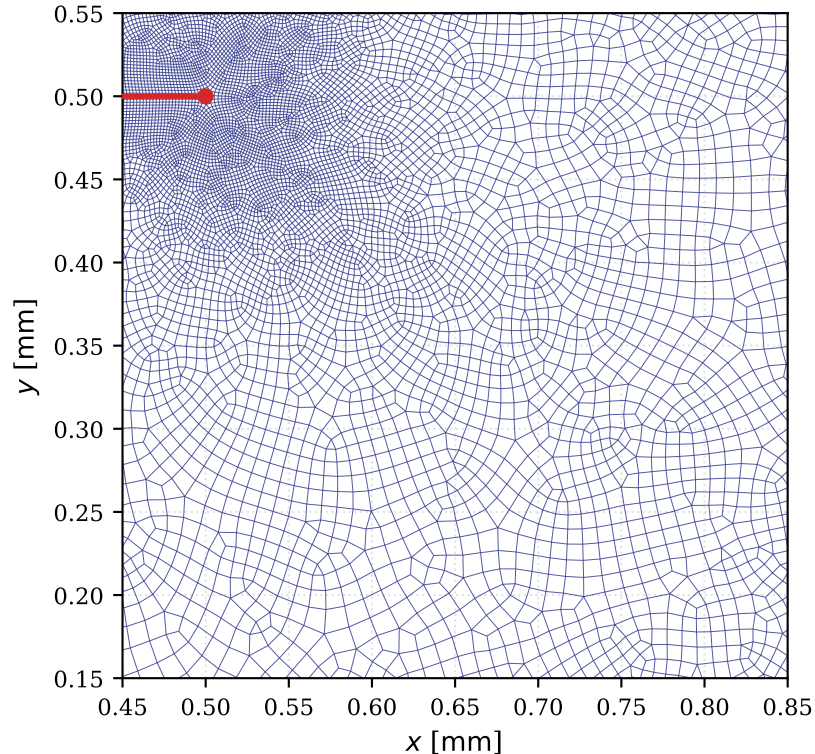
Step-2 Final Frame ($u_x = 20\mu\text{m}$)
Crack-Tip Singularity Zoom [$h_{\min} = 1.0\mu\text{m}$]



Step-1 Final Frame ($u_x = 10\mu\text{m}$)
Refinement Corridor & Shear Fan Transition Zone



Step-2 Final Frame ($u_x = 20\mu\text{m}$)
Refinement Corridor & Shear Fan Transition Zone



QUANTITATIVE MESH TOPOLOGY & PROVENANCE SUMMARY:

- Step-1 Final Frame Mesh (Active Solver Retest 1411103): 22,530 FEs (21,962 quads + 568 tris, 22,642 nodes) | SHA-256: bd02d73c2bc199db...
- Step-2 Final Frame Mesh (Alternative Candidate) : 22,405 FEs (21,827 quads + 578 tris, 22,512 nodes) | SHA-256: 9c453eb3b004c3a0...
- Element Delta: -125 FEs (-0.55%) | Node Delta: -130 nodes (-0.57%) | Sizing Rule: UNIFORM_ERROR with errorTarget = 2.0% on CPS4 pre-analysis.
- Visual Topology Comparison: Both meshes exhibit identical spatial structure – a dense refinement singularity at (0.5, 0.5) mm that fans obliquely downward along the primary shear band towards (0.81, 0.0) mm with coarse transition ($h \approx 20\mu\text{m}$) in far-field boundaries.